Matched Shot Distributions: Scientific Design and Verification

Purpose and Estimands

This extension compares straight, draw, and fade patterns with a common **physical zero-error landing point**, followed by prescribed delivery noise. No simulated endpoint or trajectory is rotated to aim at the target. Nominal face direction, path, and speed are inputs solved before sampling. Matching two endpoint coordinates does not match the entire trajectory, launch state, peak height, descent angle, or energy. Curved trajectories remain different by design.

The study estimates changes in landing spread, bias, supported-shot probability, and conditional course scoring under explicit equipment and delivery assumptions. It does not identify a universal preferred shape, a measured human variability distribution, or the performance of a particular commercial club.

This is an editable research reference, separate from the governed engineering manual. Its Standalone LaTeX Version is regenerated from this research Markdown with Pandoc, not from the engineering manual. The Astra Design Review, Phase-One Reference, issue records in issues/, executable tests, and immutable result receipts form its evidence chain. Implementation and results remain under development until their acceptance evidence is recorded.

Experimental Controls

For each club, use static loft and lie from the cited specification, fixed attack angle, and explicit assumed head properties. Pitch the complete club by forward lean of 0°, 5°, or 10°. Then solve nominal face and speed, with path constrained by face-to-path difference: 0° for straight, −1.5° for draw, +1.5° for fade; the stronger curves use ±3°. The target line remains the fixed +X direction throughout.

Two target protocols distinguish the questions:

1. **Within-Lean Matching:** Straight, draw, and fade land at the same target for a given club and lean. The target is the zero-error straight shot at that club's reference speed and lean. Distances may differ across lean settings.
2. **Across-Lean Matching:** Attempt the same physical target distance as the zero-lean straight reference. Report all infeasible conditions with their fixed bounds and reason. Reachability is not guaranteed by an optimizer's termination.

The across-lean Monte Carlo control uses the shaft-only and combined-2-mph profiles at both twist SDs and both curve strengths (72 planned cells). It holds target distance constant where matching succeeds. It is a matched-delivery lean comparison: matching also changes nominal speed, face and path, and a fixed absolute speed SD gives a different coefficient of variation at different nominal speeds. It does not identify a pure isolated lean effect or establish results for the seven other noise profiles.

Speed search bounds are fixed at 75–125% of the reference speed; nominal face bounds remain local to the intended shape. The accepted raw landing residual must be at most 0.05 m, checked independently after optimization. The frozen experiment plan records actual bounds and solved inputs. Infeasible cases remain in the matrix; neither attack angle nor static loft is silently changed to populate them.

Equipment and Provenance

Club	Static Loft	Lie at Zero Lean	Reference Speed	Attack Angle	Head Mass
Driver	10°	58.5°	45 m/s	−0.9°	0.200 kg

Club	Static Loft	Lie at Zero Lean	Reference Speed	Attack Angle	Head Mass
7-Iron	34°	63°	36 m/s	−4°	0.272 kg
Pitching Wedge	46°	64°	32 m/s	−5°	0.300 kg

The Titleist GT2 specification provides the selected driver loft/lie. The historical T100 specification provides the iron/wedge lofts and lies. These references establish geometry only. Speed, attack, mass, CG, inertia, usable face dimensions, restitution, friction, and curvature are research assumptions. The hybrid presets are **not** reconstructions or performance predictions for GT2 or T100 clubs.

Club	CG Depth	Principal Inertias: Normal, Heel, Up	Face Half-Width / Half-Height	Bulge / Roll Radius
Driver	35 mm	0.00025, 0.00040, 0.00045 kg·m²	50 / 27 mm	300 / 300 mm
7-Iron	10 mm	0.00012, 0.00020, 0.00025 kg·m²	35 / 23 mm	Flat / Flat
Pitching Wedge	8 mm	0.00010, 0.00020, 0.00022 kg·m²	35 / 25 mm	Flat / Flat

Restitution is 0.83 and friction coefficient 0.4 in the primary experiment. Inertia and CG sensitivity must accompany off-center conclusions. McNally et al. (2016) establishes the importance of CG depth, inertia, and face geometry for off-center trajectory sensitivity; it does not validate these assumed parameter values. Manufacturer loft alone is insufficient to identify gear effect.

Frames and Rigid Delivery Geometry

The native frame is right-handed: +X toward target, +Y toward the right-handed golfer/left, +Z upward. Reported lateral coordinates and face/path angles are positive right. Contact horizontal offset is positive toward the toe (−face-heel axis); vertical offset is positive toward the top of the face.

Starting with the static face normal and zero-lean shaft axis, apply a rigid rotation about +Y for forward lean, then a negative world-Z rotation for positive-right nominal face direction. Apply the stochastic shaft rotation using Rodrigues' formula about that delivered shaft axis. Positive shaft rotation closes the right-handed face. Zero twist gives delivered loft = static loft − forward lean. These rotations preserve the shaft–face angle and orthonormal face frame.

With static loft L, lie β, lean ℓ, nominal right-positive face f, and shaft twist θ, the geometric construction is

$$\mathbf{n}_0 = (\cos L, 0, \sin L)^T, \qquad \mathbf{s}_0 = (0, \cos\beta, \sin\beta)^T,$$

$$Q = R_z(-f)R_y(\ell), \quad \mathbf{s} = Q\mathbf{s}_0, \quad \mathbf{n} = R_{\mathbf{s}}(\theta)Q\mathbf{n}_0.$$

Rodrigues' rotation is $R_{\mathbf{s}}(\theta) = I\cos\theta + [\mathbf{s}]_\times \sin\theta + (1-\cos\theta)\mathbf{s}\mathbf{s}^T$. The reported face and loft are $\operatorname{atan2}(-n_y, n_x)$ and $\operatorname{atan2}(n_z, \sqrt{n_x^2 + n_y^2})$. Internal trigonometry uses radians; public angles use degrees. This explicitly explains why shaft rotation can change both face and loft and why forward lean changes that coupling. It does not establish an observed player's closure trajectory.

At zero shaft twist, the first-order sensitivities follow directly from $d\mathbf{n}/d\theta = \mathbf{s} \times \mathbf{n}$:

$$\frac{df}{d\theta} = -\frac{\sin\beta\cos L}{\cos(L-\ell)}, \qquad \frac{d\lambda}{d\theta} = -\cos\beta, \qquad \frac{d\lambda}{df} = \cot\beta\,\frac{\cos(L-\ell)}{\cos L}.$$

Here λ is delivered loft. Ratios are unchanged when all angle increments use degrees. Consequently, fixed shaft-twist SD produces approximately the same first-order loft SD across lean settings, while face-angle SD changes. This does not mean carry must increase on every closed-face miss: launch height, spin, ball speed and flight response also change. Finite-angle results and actual landing slopes are computed by the full model; these derivatives describe local geometric sensitivity only. The Independent Geometry and Carry Derivative Checks provide the analytical derivation, finite-difference evidence and club-specific limitations.

Unlike phase one's constant-delivered-loft sensitivity, forward lean here changes both face loft and shaft orientation. A 10° static driver at 10° lean can have zero delivered loft and a nonairborne nominal launch; that is a physical-domain limitation to report, not a missing value to interpolate.

Two-Body Contact Law

The previous off-center path combined a scalar effective mass with additive heuristic spin and omitted the corresponding club rotational impulse. It is unsuitable for claiming mechanically consistent strike dispersion. The new research path uses a full club inertia tensor and contact lever, without additive gear spin.

Let c be club-minus-ball relative contact velocity, n the unit outward local face normal, r_c and r_b the CG-to-contact levers, and [r]× the cross-product matrix:

$$\mathbf{c} = \mathbf{v}_c + \boldsymbol{\omega}_c \times \mathbf{r}_c - \mathbf{v}_b - \boldsymbol{\omega}_b \times \mathbf{r}_b, \qquad \mathbf{r}_b = -R\mathbf{n}.$$

The delivery adapter assumes zero initial club angular velocity and a stationary, nonspinning ball. Shaft twist is an orientation error, not a closure angular speed. Contact-generated club rotation is retained after impact. For impulse J applied to the ball:

$$K = \left(\tfrac{1}{m_c} + \tfrac{1}{m_b}\right) I - [\mathbf{r}_c]_\times I_c^{-1} [\mathbf{r}_c]_\times - [\mathbf{r}_b]_\times I_b^{-1} [\mathbf{r}_b]_\times, \qquad \mathbf{c}_{\text{after}} = \mathbf{c}_{\text{before}} - K\mathbf{J}.$$

K has units kg^{-1}, impulse J has units N·s, and relative velocity c has units m/s. The ball is a solid sphere in this idealization. The full tensor is rotated from the specified principal frame into world coordinates before the impulse calculation.

$$\Delta\mathbf{v}_b = \mathbf{J}/m_b, \quad \Delta\boldsymbol{\omega}_b = I_b^{-1}(\mathbf{r}_b \times \mathbf{J}), \quad \Delta\mathbf{v}_c = -\mathbf{J}/m_c, \quad \Delta\boldsymbol{\omega}_c = -I_c^{-1}(\mathbf{r}_c \times \mathbf{J}).$$

The primary law integrates Coulomb contact in normal-impulse coordinates. For each positive normal increment dP, solve the tangential maximum-dissipation disk problem using the current relative contact velocity and the full coupled K. Compression ends at zero normal approach velocity. Store compression work $E_c = \int c_n \, dP$; restitution returns $e_{energy}^2 E_c$ through $\int -c_n \, dP$. The coefficient 0.83 is therefore **energetic restitution**, not a prescribed normal velocity ratio in coupled off-center contact. Resolution is explicit and must pass refinement checks. The optional Numba implementation uses no fast-math transformations and is checked against the Python reference.

For an orthonormal tangent basis T and current relative velocity c, the incremental disk problem is

$$\min_{\|\Delta\mathbf{J}_t\| \le \mu\, dP} \tfrac{1}{2}\Delta\mathbf{J}_t^T (T^T K T)\Delta\mathbf{J}_t - [T^T(\mathbf{c} - dP\,K\mathbf{n})]^T \Delta\mathbf{J}_t.$$

The total increment is $dP\,\mathbf{n} + T\Delta\mathbf{J}_t$. Normal work uses trapezoidal integration, with bounded bisection at the compression and restitution endpoints. The primary resolution is 128 compression steps; source-bound refinement to 256/512 steps and flight dt 0.02/0.01/0.005 s is recorded in the Independent Numerical Evidence.

A separate maximum-dissipation/Newton comparator minimizes $½J^TKJ - c^TJ$ subject to $n^TKJ = (1+e)c_n$ and the impulse friction cone. This comparator does not claim to reproduce the full slip/grip history. An

earlier stick-first candidate passed simple centered checks but violated passivity in Astra's coupled elastic counterexample; that failed experiment motivated the corrected comparator and incremental primary law. It is not used in the final primary pathway.

All four body velocity states receive equal-and-opposite linear and angular impulse updates. Validate work recovery, cone bounds, energy passivity, linear/angular momentum, frame covariance, and numerical refinement; do not replace failed numerical solutions with heuristic launches. These rigid contact laws are not measured tangential-compliance models. Conservation and convergence establish implementation consistency, not empirical golf-ball/club accuracy. Shaft flexibility, face compliance, speed-dependent restitution, local friction, grooves, and turf interaction remain unqualified. The external measurement obligation remains open as #11984 and in the Deferred-Validation Catalog.

For curved faces, a local parabolic surface provides a declared geometry approximation. Surface displacement along the center normal is $-\frac{1}{2}(h^2/R_b + v^2/R_r)$, and its normal is proportional to $n + (h/R_b)t_{toe} + (v/R_r)t_{up}$. A zero radius explicitly selects a flat direction; it is not evaluated as a division by zero.

Stochastic Delivery and Covariance

The primitive error vector is ordered as shaft rotation (degrees), path (degrees), speed (m/s), toe/heel contact (m), high/low contact (m), forward lean (degrees), and attack angle (degrees). A multivariate Gaussian is a transparent local sensitivity assumption, not an established distribution for all human motion. Without player data there is no justified empirical tail or covariance fit.

Default comparisons include shaft-rotation SD 1° and 2°, path SD 0.5°, speed SD 1 and 2 mph (exact conversion 0.44704 m/s per mph), and strike SD 0.25 inch = 0.00635 m on **each** contact axis. The radial strike SD is not 0.25 inch. Baseline and progressively combined noise profiles separate contributions before their interactions are considered.

Users can edit all SDs and the complete symmetric correlation matrix. Require a unit diagonal, finite entries, and positive semidefiniteness. Zero-SD dimensions are exactly constant. Draw one seeded standardized sample matrix and reuse corresponding shot indices across patterns. Preserve the requested covariance and report the realized input covariance. Actual face angle is derived nonlinearly from shaft rotation and lean; a shaft/path correlation must not be labelled an exact face/path correlation. Report realized face/path covariance explicitly.

Do not clip, retry, or resample off-face, nonairborne, or otherwise unsupported deliveries. Preserve their input rows and failure categories. Supported-endpoint statistics and density are explicitly conditional when any shots are unsupported; report the valid fraction and interval alongside them. Full-population strokes gained is unavailable if an unsupported outcome has no defensible terminal lie. Conditional paired scoring uses the intersection of supported indices across all patterns, with the reduced population stated.

Density, Uncertainty, and Course Scoring

Academic figures use common axes, units, and density normalization across patterns. Show points, binned probability density, marginal distributions, and empirical 50%, 80%, and 95% highest-density regions. Report bin resolution and achieved probability mass; tied bins can exceed the requested threshold. Do not manufacture a two-dimensional Gaussian width for a nearly one-dimensional input/output manifold. Export publication PNG, SVG, and PDF; inspect rendered output for clipping, overlaps, and readable annotations.

Report lateral and longitudinal SD, full covariance, mean bias, radial RMSE and quantiles, target containment, and paired differences with sampling uncertainty. The same approximate historical scoring baseline as phase one applies: driver tee/fairway/rough; iron/wedge approach/green/rough. Landing is a

hypothetical stopping point. Roll, hazards, and wedge stopping are not inferred. Sampling intervals exclude model, equipment, and course uncertainty. Course-dimension and off-center head-parameter sensitivities remain separate from Monte Carlo error.

Pilot, Final Matrix, and Acceptance

First run small deterministic and Monte Carlo pilots. The final 10,000-shot-per-pattern matrix begins only after software contracts, physical identities, matching residuals, numerical refinement, population accounting, plot review, and an independent Astra review pass. Save unavailable cells as explicit records rather than omitting them.

Requirement	Executable Evidence	Acceptance
Static Loft and Lean	`test_club_geometry.py`, `test_matched_config.py`	Rotation identities, physical ranges, sourced static values
Correlated Noise	`test_noise.py`	PSD rejection, reproducibility, zero variance, empirical covariance
Raw Nominal Matching	`test_nominal_matching.py` and Real Pilot Receipts	Physical raw X/Y residual ≤0.05 m; fixed bounds; no aiming rotation
Off-Center Contact	Independent Rigid Contact Tests	Momentum, passive energy, restitution, frame covariance, central limit, gear direction
Unsupported Samples	Matched Runner Tests	Every input retained; no resampling; explicit conditional estimands
Density Figures	Density/Plot Tests and Rendered Review	Normalization, ties, degeneracy, common scale, readable exports
Final Results	Source-Bound Manifests and Independent Recalculation	Exact reproducibility, complete matrix accounting, no stale provenance

Final reports must link exact source revisions, native binary identity, input specifications, seeds, solver bounds, numerical tolerances, result hashes, pilot evidence, and review dispositions. Historical phase-one outputs remain labelled with their original model and controls.